1. Besh xonali $\overline{2x78y}$ son 72 ga qoldiqsiz bo'linsa, $xy - 3y$ ni hisoblang.
A) 20
B) 15
C) 12
D) 10

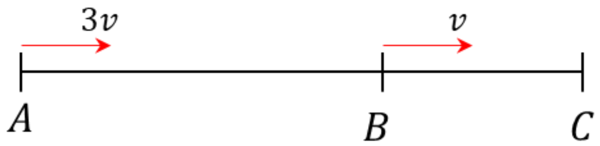
2. Hisoblang.

$$\frac{0{,}429}{0{,}03} + \frac{0{,}128}{0{,}08} + \frac{0{,}0096}{0{,}012}$$

A) 1,67
B) 16,7
C) 2,39
D) 23,9

3. Tarkibida 30% tuz bo'lgan 16 kg eritmani tarkibida 50% tuz bo'lgan eritma bilan aralashtirildi. Hosil bo'lgan eritmaning tarkibida 42% tuz bo'lsa, ikkinchi eritmadan necha kg olingan?
A) 24
B) 30
C) 22
D) 50

4. A va B shaharlar orasidagi masofa, B va C shaharlar orasidagi masofadan ikki marta katta. Avtomobil A shahardan B shahargacha shahargacha $3v$ tezlik bilan, B shahardan C shahargacha v tezlik bilan yurib, jami yo'l uchun 1 soat ketkizdi. Avtomobil A dan B gacha bo'lgan yo'lni necha daqiqada bosib o'tgan?



A) 24
B) 30
C) 36
D) 40

5. Ifodani soddalashtiring.

$$\frac{(3^k + 7 \cdot 3^k)^3}{(9^k + 9 \cdot 9^k)^2}$$

A) $\frac{10}{3^{2k}}$
B) $6{,}4 \cdot 3^{-k}$
C) $10{,}24 \cdot 3^{-3k}$
D) $5{,}12 \cdot 3^{-k}$

6. Hisoblang: $\frac{1}{\sqrt{7-\sqrt{24}}+1} - \frac{1}{\sqrt{7+\sqrt{24}}-1}$
A) 1
B) $2\sqrt{6}$
C) 0
D) $-2\sqrt{6}$

7. $\sqrt{23 - 8\sqrt{7}} \cdot \sqrt{\left(2 - \sqrt{7}\right)^2}$ hisoblang.
A) $15 - 6\sqrt{7}$
B) $2\sqrt{7} - 1$
C) $6\sqrt{7} - 15$
D) $2\sqrt{7} + 1$

8. $x^3 - 9x^2 + mx - 6 = 0$ tenglamaning ildizlari arifmetik progressiyani tashkil etsa, m ni toping.
A) 3
B) 5
C) 10
D) 20

9. $a_n = \frac{2}{3^{n-1}}$ geometrik progressiya umumiy hadi bo'lsa, hadlar yig'indisini toping?
A) $\frac{1}{3}$
B) $\frac{3}{2}$
C) 3
D) $\frac{9}{2}$

10. x, y butun musbat sonlar $2x \neq y$ va $2x^3 - yx^2 - 2xy^2 + y^3 = 10x - 5y$ tenglik o'rinli bo'lsa, $x + y$ ni hisoblang.
A) 1
B) 2
C) 5
D) 4

11. Agar $a = 7{,}5$ va $b = 6{,}5$ ga teng bo'lsa, $\left(\frac{2ab}{a^2-b^2} + \frac{a-b}{a+b}\right) : \frac{2a}{a+b} + \frac{b}{b-a}$ ni toping.
A) $-\frac{1}{13}$
B) $\frac{1}{13}$
C) $\frac{1}{15}$
D) $-\frac{1}{15}$

12. Hisoblang: $2\sin^2(3°45') \cdot (1 + \cos(7°30')) - (2\sin^2(3°45') - 1)^2$
A) $\frac{\sqrt{6}+\sqrt{2}}{4}$
B) $-\frac{\sqrt{6}-\sqrt{2}}{4}$
C) $-\frac{\sqrt{6}+\sqrt{2}}{4}$
D) $-\frac{\sqrt{2}}{2}$

13. $\sin x - \sqrt{3}\cos x = \sqrt{3}$ tenglamaning eng katta manfiy ildizini toping.
A) $-\frac{\pi}{2}$
B) $-\pi$
C) $-\frac{3\pi}{2}$
D) -2π

14. Tengsizlikni yeching: $3 \cdot 4^x + 2 \cdot 9^x - 5 \cdot 6^x < 0$
A) $(1; \infty)$
B) $(0; 1)$
C) $(0; \infty)$
D) $(0; 4)$

15. $\log_3(5 + \log_2(3x - 2)) < 2$ tengsizlikning eng katta qiymatini toping?
A) 1
B) 2
C) 3
D) 5

16. $|3x + 9| = 15$ tenglamaning haqiqiy ildizlari ko'paytmasini toping.(Agar u bitta bo'lsa, shu ildizini toping)
A) -8
B) 16
C) -16
D) 2

17. $2^{3x} \cdot 3^{x} = 576$ tenglamaning ildizini toping.
A) 2
B) 3
C) 4
D) 5

18. $|x^2 + 2x + 7| < |x^2 - 3x + 22|$ tengsizlikning natural qiymatlari nechta?
A) 3
B) 4
C) 2
D) 5

19. $\frac{\sqrt{12-x-x^2}}{2x-7} \leq \frac{\sqrt{12-x-x^2}}{x-5}$ tengsizlikning eng katta butun yechimini toping.
A) 1
B) 2
C) 3
D) 4

20. $f(x) = 11x^4 + 7 + \sqrt{\log_8(x-8)} + \frac{x^4}{x^2-16}$ funksiyaning aniqlanish sohasini toping?
A) $(2;\infty)$
B) $[4;\infty)$
C) $[9;\infty)$
D) $(2;4) \cup (4;\infty)$

21. Integralni hisoblang:

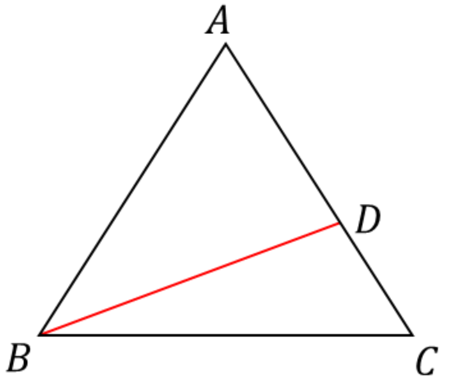
$$\int \frac{x^2+4x+3}{x^2+5x+4}dx$$

A) $x+\ln|x+4|+C$
B) $x+\ln|x+3|+C$
C) $x-\ln|x+4|+C$
D) $x-\ln|x+3|+C$

22. $f(x)=x^2+2x-1$ bo'lsa, $f'(0)\cdot f'(1)$ ni hisoblang.
A) 8
B) 9
C) 10
D) 11

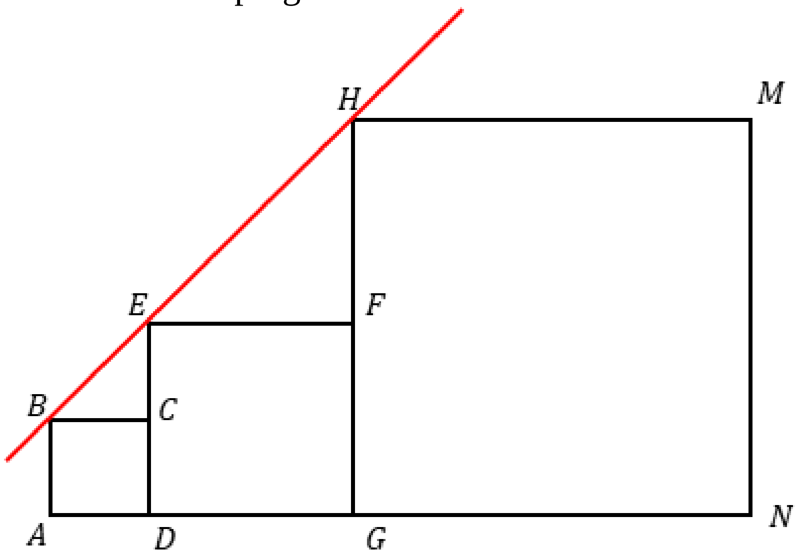
23. Yuzasi 30 ga teng bo'lgan ABC uchburchakning AC tomonida D nuqta shunday olinganki, $AD: DC = 2: 3$ tenglik o'rinli. D nuqtadan BC tomonga uzunligi 9 ga teng bo'lgan DE perpendikulyar o'tkazilgan. BC tomon uzunligini toping.
A) 3
B) 5
C) 4
D) 6

24. ABC teng tomonli uchburchak, $|AB|=6$, $\frac{S_{ABD}}{S_{ABC}}=\frac{1}{3}$ bo'lsa, $|BD|$ ni toping?



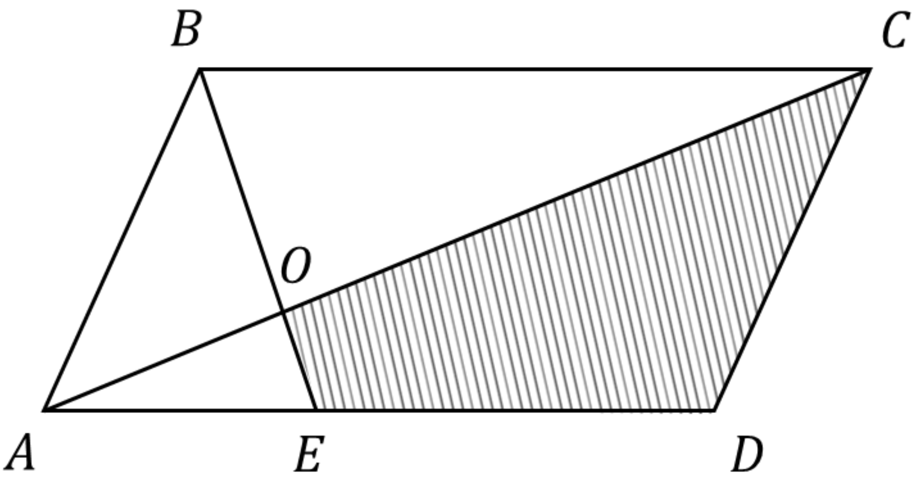
A) $\sqrt{10}$ B) $2\sqrt{7}$ C) $\sqrt{3}$ D) $2\sqrt{3}$

25. Rasmda B, E va H uchlari bir to'g'ri chiziqda yotuvchi $ABCD$, $DEFG$ va $GHMN$ kvadratlar taasvirlangan. $ABCD$ va $DEFG$ kvadratlar tomonlari mos ravishda 1 va 3 ga teng bo'lsa, $GHMN$ kvadrat tomonini toping.



A) 6
B) 7
C) 8
D) 9

26. $ABCD$ parallelogramda AD tomonda E nuqta tanlab olindi. AC diagonal bilan BE kesam O nuqtada kesishadi. Agar ABO uchburchak yuzasi 12 ga, AOE uchburchak yuzasi 4 ga teng bo'lsa, $COED$ to'rtburchak yuzasini toping.



A) 36
B) 40
C) 44
D) 48

27. $\sqrt{x^2-14x+49}+\sqrt{x^2+4x+4}$ ifodaning eng kichik qiymatini toping.
A) 7
B) 9
C) $\sqrt{53}$
D) 10

28. Aylana $ABCDEFGH$ sakkizburchakka ichki chizilgan. Agar $AB=5$, $BC=CD=4$, $DE=2$, $EF=6$, $FG=7$ bo'lsa, $GH-HA$ ni toping?
A) 2
B) aniqlab bo'lmaydi
C) -2
D) 28

29. $ABCD$ parallelogram yuzasi 36 ga teng. $\overrightarrow{BE}=\frac{1}{2}\overrightarrow{BC}$ bo'lsa, $ABED$ to'rtburchak yuzini toping.
A) 54
B) 45
C) 27
D) 18

30. Tekislikdan 4 masofada yotgan nuqtadan ikkita teng og'ma o'tkazilgan. Agar og'malar o'zaro perpendikulyar va tekislikka o'tkazilgan perpendikulyar bilan $30°$ ga teng burchaklar tashkil etishi ma'lum bo'lsa, og'malarning asoslari orasidagi masofani toping?
A) $\frac{8\sqrt{6}}{3}$
B) $\frac{8\sqrt{3}}{3}$
C) $\frac{4\sqrt{6}}{3}$
D) $\frac{4\sqrt{3}}{3}$

31. $A = \{(x;y)|x^2 + y^2 = 4, x, y \in R\}$, $B = \{(x;y)|x - y = 2, x, y \in R\}$ bo'lsa, $A \cap B$ to'plamning qism to'plamlar sonini toping.
A) 2
B) 4
C) 8
D) 16

32. To'rt xonali sonlar ichidan raqamlari juft sonlardan iborat bo'lgan va 5 ga karrali son bo'lish ehtimolligini toping.
A) $\frac{1}{9}$
B) $\frac{1}{72}$
C) $\frac{1}{45}$
D) $\frac{1}{90}$

<table>
<tr><td colspan="2">Topshiriqlar(33-35) va javob variantlari (A-F) ni o'zaro moslashtiring.

Ustada bir nechta balandligi asosining diametridan ikki marta katta bo'lgan bir xil silindr shakldagi g'o'lalar bor. ($\pi \approx 3$ deb oling)</td></tr>
<tr><td>33. Usta bu go'ladan iloji boricha eng katta hajmli shar o'yib olmoqchi. Bunda shar hajmining g'o'la hajmiga nisbatini toping.

34. Usta bu go'ladan eng katta hajmli parallelepiped shaklidagi to'sin. yasamoqchi. O'yib olingan chiqindi hajmining to'sin hajmiga nisbatini toping.

35. Usta bu go'ladan eng katta hajmli konus yasamoqchi. O'yib olingan chiqindi hajmining g'o'la hajmiga nisbatini topíng.</td><td>A) $\frac{1}{2}$

B) 1

C) $\frac{1}{3}$

D) $\frac{2}{3}$

E) $\frac{1}{4}$

F) $\frac{3}{4}$</td></tr>
</table>

36. $\begin{cases} x^2+y^2-2x+3y-9=0 \\ 2x^2+2y^2+x-5y-1=0 \end{cases}$

a) Tenglamalar sistemasini qanoatlantiruvchi nechta $(x;y)$ juftligi mavjud?

Javob a)____________________________________

b) Yechimlari $(x_1;y_1);(x_2;y_2);\ldots(x_n;y_n)$ bo'lsa, $x_1+y_1+x_2+y_2+\cdots x_n+y_n$ ni toping

Javob b)____________________________________

37. $(4\cos^2 x-1)\sqrt{4\pi^2-x^2}=0$ tenglama berilgan.

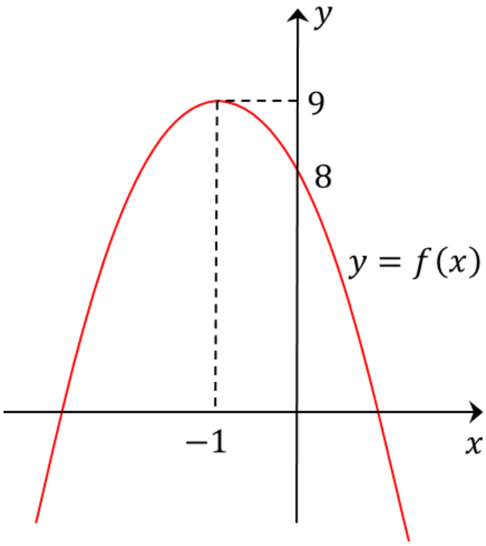
a) Tenglamaning nechta haqiqiy ildizi bor?

Javob a)____________________________________

b) Tenglamaning $[0;2\pi]$ oraliqdagi ildizlar yig'indisini hisoblang.

Javob b)____________________________________

38. Rasmda $y = ax^2 + bx + c$ parabola grafigi tasvirlangan.



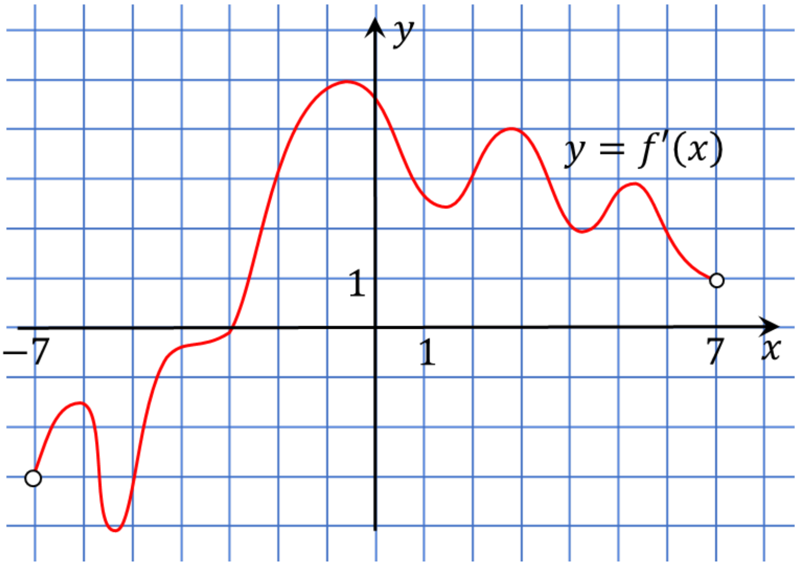
a) $a + b + c$ ni hisoblang.

Javob a)___________________________________

b) $f(x)$ va OX o'qi bilan chegaralangan soha yuzini toping

Javob b)___________________________________

39. Rasmda $y = f(x)$ funksiyaning hosilasi grafigi tasvirlangan.



a) $f(x)$ funksiyaning hosilasining nechta lokal maksimum nuqtasi mavjud?

Javob a)___________________________________

b) $f(x)$ funksiyaning nechta lokal minimum nuqtasi mavjud?

Javob b)___________________________________

40. $f(n) = e^2 - 1$ va $f(m) = 0$ bo'lsa,

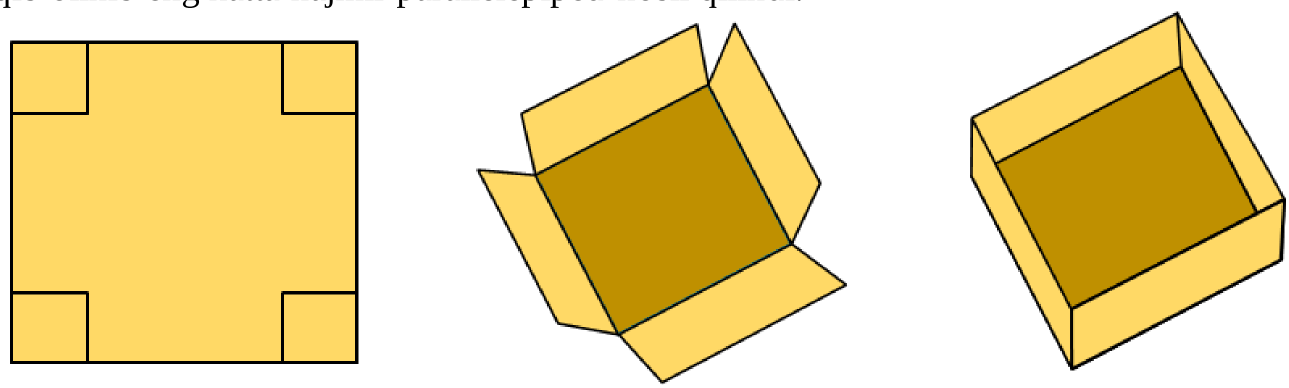
a) $A = \frac{\pi}{4}\int_m^n \frac{4f'(x)}{1+f(x)}dx$ integralni toping.

Javob a)___________________________________

b) Markaziy burchagi α, radiusi 1 ga teng bo'lgan yoyning uzunligi A ga teng bo'lsa α ni toping,

Javob b)___________________________________

41. $ABCD$ kvadratning tomoni 18 ga teng. Rasmda kvadratning har bir uchidan kvadratchalar qirqib olinib eng katta hajmli parallelepiped hosil qilindi.



a) Eng katta hajmli parallelepipedning to'la sirtini toping.
Javob a)___________________________________

b) Parallelipiped hajmini toping.
Javob b)___________________________________

42. Markazi $O(3;7)$ nuqtada bo'lgan aylana berilgan. Bu aylanaga $(a;-4)$ nuqtadan urinma o'tkazilgan. Bu urinmaning burchak koeffitsiyenti $\frac{5}{4}$ ga teng bo'lsa,

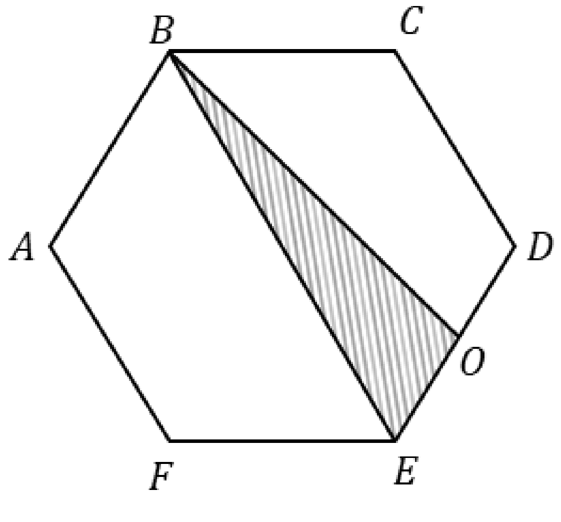
a) a ni toping.

Javob a)____________________________________

b) Aylana radiusini toping

Javob b)____________________________________

43. $ABCDEF$ muntazam otiburchakning tomoni $\sqrt{2\sqrt{3}+3}$. FBC burchak bissektrisasi DE tomonni O nuqtada kesib o'tsa,



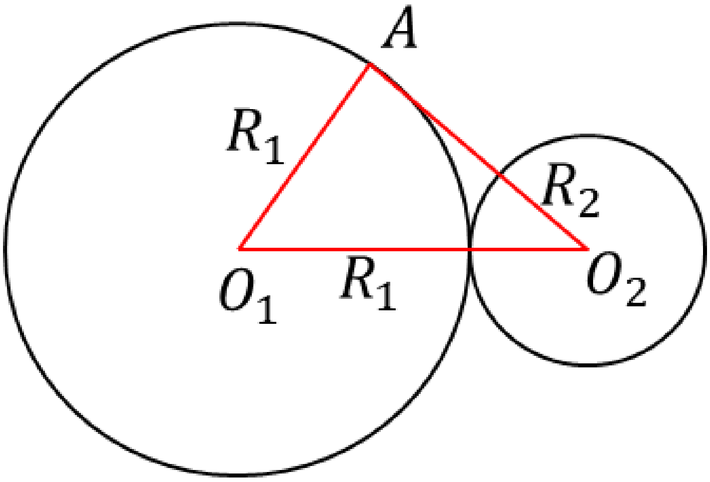
a) $|BO|$ kesma uzunligini toping.

Javob a)____________________________________

b) BOE uchburchak yuzini toping.

Javob b)____________________________________

44. Ikkita doira tashqi ravishta urinadi, O_1 va O_2 lar mos ravishda katta va kichik doiralar markazlari. R_1 va R_2 lar mos ravishda katta va kichik doiralar radiuslari, O_2A katta doiraga urinma bo'lib va kichik doira radiusidan 3 marta katta bo'lsa



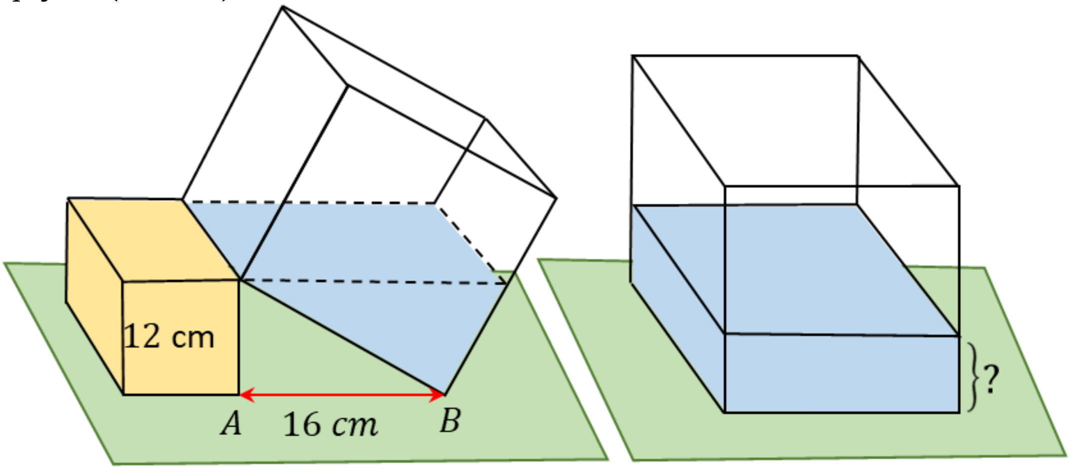
a) Katta doira uzunligining kichik doira uzunligiga nisbatini toping.

Javob a)__________________________________

b) Katta doira yuzining kichik doira yuziga nisbatini toping.

Javob b)__________________________________

45. Balandligi 12 sm bo'lgan to'g'ri burchakli paralleliped shaklidagi yog'ochga kub shaklidagi idish rasmdagidek qiya qo'yildi, so'ngra idishga yog'och baladliga teng suv quyildi(1-rasm).



$1 - rasm$ $2 - rasm$

a) Ushbu yog'och bo'lagi olinsa idish 2-rasmdagi holatga keladi. 2-idishdagi suv balandligini toping. (cm)

Javob a)__________________________________

b) 2-rasmdagi idishga eng kamida necha litr suv quyilsa, idish to'ladi?

Javob b)__________________________________